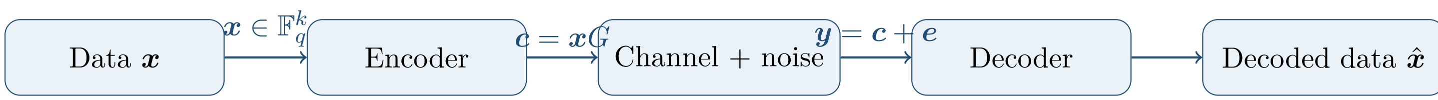
Coding Theory: Linear Codes

Lecture Notes in LaTeX



Scope of these notes. These notes reorganize the “Linear Codes” slide deck into a continuous lecture-note format. They cover finite fields, linear encodings, (n, k) linear codes, generator matrices, systematic form, repetition and parity-check codes, and parity-check matrices.

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Contents

1. What is coding theory?

Coding theory studies how to transmit or store information reliably in the presence of corruption. In a communication setting, a sender starts with information symbols, transforms them into an encoded word, sends that word through a noisy channel, and the receiver tries to recover the original data from the corrupted output.

Key idea

The central principle is redundancy: we intentionally add structured extra symbols so that the receiver can detect or correct corruption.

A generic communication model has the form

$$\boldsymbol{x} = (x_1, \dots, x_k) \xrightarrow{\text{encoder}} \boldsymbol{c} = (c_1, \dots, c_n) \xrightarrow{\text{noise } \boldsymbol{e}} \boldsymbol{y} = \boldsymbol{c} + \boldsymbol{e} \xrightarrow{\text{decoder}} \hat{\boldsymbol{x}} = (\hat{x}_1, \dots, \hat{x}_k),$$

where usually $n \geq k$. The vector $\boldsymbol{c}$ is called a *codeword*.

1.1 Encoding

Let A be an alphabet. An *encoding* is a map

$$A^k \longrightarrow A^n, \qquad (x_1, \dots, x_k) \longmapsto (c_1, \dots, c_n),$$

with $n \geq k$. The output vector $\boldsymbol{c} = (c_1, \dots, c_n)$ is the transmitted codeword.

Example 1.1 (A first encoding)**.** *When $k = 1$, a very simple encoding repeats the same symbol:*

$$(x_1) \longmapsto (x_1, x_1, \dots, x_1) \in A^n.$$

If the alphabet is binary, $A = \{0, 1\}$, then

$$(0) \mapsto (0, 0, \dots, 0), \qquad (1) \mapsto (1, 1, \dots, 1).$$

This is called the repetition code*.*

2. Discrete alphabets and finite fields

In linear coding theory, alphabets are usually finite fields. The first examples come from arithmetic modulo a prime.

2.1 Arithmetic modulo p

For a prime number p, arithmetic modulo p means that every computation is reduced by taking its remainder upon division by p. The resulting set is

$$\mathbb{F}_p = \{0, 1, \dots, p - 1\}.$$

Addition and multiplication modulo 2

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Addition and multiplication modulo 3

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

2.2 What goes wrong modulo a composite number?

If we work modulo a non-prime integer, arithmetic no longer behaves like field arithmetic.

For example, modulo 4:

$$2 \cdot 2 \equiv 0 \pmod 4.$$

This has several consequences:

- from $2x = 0$, we cannot conclude $x = 0$, because $x = 2$ also works;
- a degree-1 equation may have more than one solution;
- two nonzero numbers can multiply to zero, so *zero divisors* appear;
- equations such as $2x = 1$ can have no solution at all.

Exercise 2.1. *Prove that if m is not prime, then integers modulo m do not form a field.*

Solution

If m is composite, then $m = ab$ for some integers a, b with $1 < a < m$ and $1 < b < m$. Modulo m, we have

$$ab \equiv 0 \pmod m.$$

Hence the class of a is nonzero and the class of b is nonzero, but their product is zero. In a field, this cannot happen: nonzero elements must be invertible, and a product of two nonzero elements cannot be zero. Therefore $\mathbb{Z}/m\mathbb{Z}$ is not a field when m is composite.

2.3 The finite field $\mathbb{F}_p$

Definition 2.2. *For a prime p, the set of integers modulo p, denoted $\mathbb{F}_p$, is a* finite field. *This means:*

- *it has exactly p elements;*

- *addition, subtraction, and multiplication are defined and satisfy the usual commutative laws;*
- *every nonzero element has a multiplicative inverse.*

Example 2.3 (Inverse modulo 5)**.** *In $\mathbb{F}_5$, the inverse of 3 is 2 because*

$$3 \cdot 2 = 6 \equiv 1 \pmod 5.$$

Example 2.4 (Elements that are their own inverse modulo 7)**.** *In $\mathbb{F}_7$, the elements 1 and 6 are self-inverse because*

$$1 \cdot 1 \equiv 1 \pmod 7, \qquad 6 \cdot 6 = 36 \equiv 1 \pmod 7.$$

2.4 Another finite field: $\mathbb{F}_4$

Finite fields do not only exist when the number of elements is prime. There is also a field with four elements. One way to describe it is to introduce an element ω satisfying

$$\omega^2 + \omega + 1 = 0 \quad \text{over } \mathbb{F}_2.$$

Equivalently,

$$\omega^2 = \omega + 1, \qquad \omega^3 = 1.$$

Then

$$\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}.$$

Its addition and multiplication tables are:

$+$	0	1	ω	ω^2
0	0	1	ω	ω^2
1	1	0	ω^2	ω
ω	ω	ω^2	0	1
ω^2	ω^2	ω	1	0

$\cdot$	0	1	ω	ω^2
0	0	0	0	0
1	0	1	ω	ω^2
ω	0	ω	ω^2	1
ω^2	0	ω^2	1	ω

From now on we write $\mathbb{F}_q$ for a finite field with q elements. Typical examples are $\mathbb{F}_p$ for prime p, and also extension fields such as $\mathbb{F}_4$.

3. Linear encoding and linear codes

Once the alphabet is a finite field $\mathbb{F}_q$, the ambient space $\mathbb{F}_q^n$ becomes a vector space. This allows us to use linear algebra to design codes.

3.1 Linear encoding

Definition 3.1. *A* linear encoding *over $\mathbb{F}_q$ is a linear map*

$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \qquad \boldsymbol{x} \longmapsto \boldsymbol{c}.$$

Because the map is linear:

- the sum of two codewords is again a codeword;
- any scalar multiple of a codeword is again a codeword.

Definition 3.2. *A subset $\mathcal{C} \subseteq \mathbb{F}_q^n$ is called a* linear code *if it is a linear subspace of $\mathbb{F}_q^n$.*

If $\mathcal{C}$ is a linear subspace of dimension k, then $\mathcal{C}$ is called an (n, k) linear code over $\mathbb{F}_q$. The integer n is the *length*, and k is the *dimension*.

3.2 Basic properties

If $\mathcal{C}$ is a linear code, then automatically:

- the zero codeword $\mathbf{0}$ belongs to $\mathcal{C}$;
- whenever $\boldsymbol{c} \in \mathcal{C}$, also $-\boldsymbol{c} \in \mathcal{C}$;
- every codeword is a linear combination of basis vectors of $\mathcal{C}$.

Proposition 3.3. *If $\mathcal{C}$ is an (n, k) linear code over $\mathbb{F}_q$, then*

$$|\mathcal{C}| = q^k.$$

Proof. Choose a basis $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_k$ of $\mathcal{C}$. Every codeword can be written uniquely as

$$x_1\boldsymbol{b}_1 + \cdots + x_k\boldsymbol{b}_k, \qquad x_i \in \mathbb{F}_q.$$

Each coefficient x_i has q choices, and there are k independent coefficients. Therefore the number of codewords is q^k. □

4. Generator matrices

Linear maps between finite-dimensional vector spaces are represented by matrices. In coding theory, the matrix representing the encoding map is called a generator matrix.

Definition 4.1. *Let $\mathcal{C}$ be an (n, k) linear code over $\mathbb{F}_q$. A* generator matrix *for $\mathcal{C}$ is a $k \times n$ matrix G whose rows form a basis of $\mathcal{C}$.*

If $\boldsymbol{x} = (x_1, \ldots, x_k) \in \mathbb{F}_q^k$, then encoding is given by

$$\boldsymbol{c} = \boldsymbol{x}G.$$

4.1 Systematic form

A particularly convenient generator matrix has the form

$$G = [I_k \mid A],$$

where I_k is the $k \times k$ identity matrix and A is a $k \times (n - k)$ matrix.

Definition 4.2. *A code is said to be in* systematic form *if it has a generator matrix of the form* $G = [I_k \mid A]$.

In this case, the first k coordinates of the codeword are exactly the information symbols. If

$$\boldsymbol{x} = (x_1, \ldots, x_k),$$

then

$$\boldsymbol{x}G = (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n).$$

So the information symbols appear explicitly in the transmitted codeword, followed by the parity symbols.

4.2 Example: repetition code

The $(n, 1)$ repetition code sends

$$(x_1) \longmapsto (x_1, x_1, \ldots, x_1).$$

Its generator matrix is simply

$$G = [1\ 1\ \cdots\ 1].$$

Indeed,

$$(x_1)G = (x_1, x_1, \ldots, x_1).$$

4.3 Example: single parity-check code

The $(n, n-1)$ single parity-check code is defined by

$$(x_1, \ldots, x_k) \longmapsto (x_1, \ldots, x_k, x_1 + \cdots + x_k),$$

where $k = n - 1$.

A generator matrix in systematic form is

$$G = \begin{bmatrix} 1 & 0 & \cdots & 0 & 1 \\ 0 & 1 & \cdots & 0 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 1 \end{bmatrix} = [I_k \mid \mathbf{1}],$$

where the last column consists entirely of ones.

Multiplying gives

$$(x_1, \ldots, x_k)G = (x_1, \ldots, x_k, x_1 + \cdots + x_k).$$

Key idea

In systematic form, the generator matrix shows exactly how redundancy is added to the original data.

5. Worked exercises on linearity

5.1 A codebook that is not linear

Exercise 5.1. *Consider the codebook of length* 4 *over* $\mathbb{F}_2$*:*

$$\{(1,0,0,0),\ (0,1,0,1),\ (1,1,0,1),\ (0,0,0,1)\}.$$

Is this a linear code? If so, provide a generator matrix.

Solution

This codebook is *not* linear.
First, a linear code must contain the zero vector $(0,0,0,0)$, but this vector is not in the codebook.
Second, linear codes must be closed under addition. In $\mathbb{F}_2$,

$$(1,1,0,1) + (0,0,0,1) = (1,1,0,0),$$

and this vector also does not belong to the codebook. Therefore the set is not a linear code.

5.2 A codebook that is linear

Exercise 5.2. *Consider the following codebook of length* 5 *over* $\mathbb{F}_2$*:*

$$\begin{gathered}(0,0,0,0,0),\ (1,0,0,1,0),\ (0,1,0,1,1),\ (1,1,0,0,1),\\ (0,0,1,0,1),\ (1,0,1,1,1),\ (0,1,1,1,0),\ (1,1,1,0,0).\end{gathered}$$

Is this a linear code? If so, provide a generator matrix.

Solution

Yes, this is a linear code. Notice that the first three coordinates run through all vectors of $\mathbb{F}_2^3$:

$$(0,0,0),(1,0,0),(0,1,0),(1,1,0),(0,0,1),(1,0,1),(0,1,1),(1,1,1).$$

This strongly suggests a systematic generator matrix of the form

$$G = \begin{bmatrix} 1 & 0 & 0 & a_{11} & a_{12} \\ 0 & 1 & 0 & a_{21} & a_{22} \\ 0 & 0 & 1 & a_{31} & a_{32} \end{bmatrix}.$$

We read off the last two coordinates from the codewords corresponding to the basis vectors:

$$(1,0,0) \mapsto (1,0,0,1,0),$$

so the first row is $(1,0,0,1,0)$;

$$(0,1,0) \mapsto (0,1,0,1,1),$$

so the second row is $(0,1,0,1,1)$;

$$(0,0,1) \mapsto (0,0,1,0,1),$$

so the third row is $(0,0,1,0,1)$.
Hence a generator matrix is

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}.$$

Now every codeword is of the form $\boldsymbol{x}G$ with $\boldsymbol{x} \in \mathbb{F}_2^3$, so the codebook is indeed linear.

6. Parity-check matrices

A linear code can be described either by its generators or by the linear equations that all codewords satisfy.

Definition 6.1. *Let $\mathcal{C}$ be an (n,k) linear code over $\mathbb{F}_q$. A matrix H of size $(n-k)\times n$ is called a* parity-check matrix *for $\mathcal{C}$ if*

$$\mathcal{C} = \{\boldsymbol{v} \in \mathbb{F}_q^n : H\boldsymbol{v}^\mathsf{T} = 0\}.$$

Thus codewords are exactly the vectors satisfying a system of $n-k$ homogeneous linear equations.

6.1 Parity-check matrix from a systematic generator matrix

Suppose

$$G = [I_k \mid A]$$

is a generator matrix in systematic form, where A is a $k \times (n-k)$ matrix. Then define

$$H = [-A^{\mathsf{T}} \mid I_{n-k}].$$

Proposition 6.2. *If $G = [I_k \mid A]$, then $H = [-A^{\mathsf{T}} \mid I_{n-k}]$ is a parity-check matrix for the code generated by G.*

Proof. First compute

$$G^{\mathsf{T}} = \begin{bmatrix} I_k \\ A^{\mathsf{T}} \end{bmatrix}.$$

Therefore

$$HG^{\mathsf{T}} = [-A^{\mathsf{T}} \mid I_{n-k}] \begin{bmatrix} I_k \\ A^{\mathsf{T}} \end{bmatrix} = -A^{\mathsf{T}} + A^{\mathsf{T}} = 0.$$

If $\boldsymbol{c} \in \mathcal{C}$, then $\boldsymbol{c} = \boldsymbol{x}G$ for some $\boldsymbol{x} \in \mathbb{F}_q^k$. Taking transposes,

$$\boldsymbol{c}^{\mathsf{T}} = G^{\mathsf{T}}\boldsymbol{x}^{\mathsf{T}}.$$

Hence

$$H\boldsymbol{c}^{\mathsf{T}} = HG^{\mathsf{T}}\boldsymbol{x}^{\mathsf{T}} = 0.$$

So every codeword lies in the kernel of the linear map $\boldsymbol{v} \mapsto H\boldsymbol{v}^{\mathsf{T}}$.

Now H has rank $n-k$ because its last $n-k$ columns form the identity matrix. Therefore its kernel has dimension

$$n - (n-k) = k.$$

But $\mathcal{C}$ is already a k-dimensional subspace contained in this kernel, so they must be equal:

$$\mathcal{C} = \ker(H).$$

Thus H is a parity-check matrix for $\mathcal{C}$. $\square$

6.2 Example: parity-check matrix of the repetition code

For the $(n, 1)$ repetition code, the generator matrix in systematic form is

$$G = [1 \mid 1\ 1\ \cdots\ 1].$$

Hence A is the row vector of all ones, so a parity-check matrix is

$$H = [-A^{\mathsf{T}} \mid I_{n-1}] = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ -1 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \cdots & 1 \end{bmatrix}.$$

Over $\mathbb{F}_2$, we have $-1 = 1$, so this becomes

$$H = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 1 \end{bmatrix}.$$

These equations force all coordinates to be equal, exactly as expected for the repetition code.

7. Final summary

Main ideas from this lecture.

- A code adds structured redundancy so that corrupted data can still be recovered.
- For linear coding theory, the alphabet is usually a finite field $\mathbb{F}_q$.
- A linear (n, k) code is a k-dimensional subspace of $\mathbb{F}_q^n$.
- Such a code contains exactly q^k codewords.
- A generator matrix G describes encoding by $\boldsymbol{c} = \boldsymbol{x}G$.
- In systematic form, $G = [I_k \mid A]$, so the data symbols appear directly in the codeword.
- A parity-check matrix H describes the code as the solution space of linear equations:

$$\mathcal{C} = \{\boldsymbol{v} \in \mathbb{F}_q^n : H\boldsymbol{v}^\mathsf{T} = 0\}.$$

- If $G = [I_k \mid A]$, then a compatible parity-check matrix is $H = [-A^\mathsf{T} \mid I_{n-k}]$.